

Effectiveness of CRISPR/Cas-9 in Inducing the CCR5 Δ 32 Mutation in Patients with HIV-1

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Abstract:

Human Immunodeficiency Virus Type 1(HIV-1) is a virus that attacks the human immune system, primarily targeting CD4⁺T-cells. Previous studies investigated the transplantation of HSCs with the CCR5Δ32 mutation, which provides natural defense against HIV-1, in patients. This study aims to evaluate the effectiveness of Clustered Regularly Interspaced Short Palindromic Repeats(CRISPR-Cas9) gene-editing technology in introducing the CCR5Δ32 mutation in stem cells using murine models. Experimental mice will be engrafted with human hematopoietic stem cells(HSCs) that have been genetically edited to contain the CCR5Δ32 mutation, and then infected with HIV-1. Conversely, the control group will not receive the mutation. Successful outcomes will demonstrate lower levels of HIV-1 viral RNA and infected CD4⁺T-cells in the experimental group, indicating viral control. The findings of this study provide a pathway towards a functional cure for HIV-1 patients through autologous stem cell transplant, potentially reducing the need for lifelong antiretroviral therapy through scalable, targeted, and non-invasive treatments.

Keywords:

HIV-1, CRISPR-Cas9, CCR5Δ32, humanized mice, human hematopoietic stem cells

Introduction:

HIV-1 remains one of the world's most pressing global health challenges. It compromises the immune system, increasing vulnerability to infections and diseases. While Highly Active Antiretroviral Therapy (HAART) can suppress viral replication, it does not eliminate viral reservoirs and requires lifelong adherence, often leading to drug resistance.¹ An alternative strategy uses CRISPR-Cas9 to target the CCR5 co-receptor, a protein on the surface of CD4⁺T-cells that HIV-1 frequently uses for cellular entry.² Individuals with a naturally occurring

homozygous CCR5 Δ 32 mutation, where a 32-base-pair deletion occurs in the CCR5 coding region, demonstrate resistance to HIV-1 or delayed disease progression because T-cells lack the CCR5 co-receptor.³ Transplanting HSCs from CCR5 Δ 32 donors into HIV-positive patients resulted in long-term remission in isolated cases like the London and Berlin Patients, but remains invasive and impractical for broad application.⁴ Recent advances in CRISPR-Cas9 gene-editing technology offer a promising solution, enabling targeted introduction of the CCR5 Δ 32 mutation into patients' HSCs through autologous stem cell transplant.^{5,2} This study aims to evaluate whether using CRISPR-Cas9 to introduce the mutation into human HSCs in mice can confer similar protection in patients naturally having CCR5 Δ 32. The efficiency of CRISPR-Cas9, indicated by reduced viral replication and eradication of infected T-cells, could reveal the feasibility of gene-editing-based approaches as a functional cure for HIV-1 in general applications, addressing the literature gap.^{6,7}

Hypothesis:

If a homozygous CCR5 Δ 32 mutation is introduced into mice using genetically edited HSCs, HIV-1 will be unable to infect CD4⁺T-cells. This decreases the levels of HIV-1 viral RNA and the frequency of infected CD4⁺T-cells, indicating resistance to HIV-1.

Rationale:

While cases like the Berlin and London patients showed success using bone marrow donors with the CCR5 Δ 32 mutation, applying CRISPR-Cas9 directly in patients through autologous stem cell therapy remains novel.^{4,5} Earlier studies explored the feasibility of CRISPR-mediated edits in vitro; however, few directly correlated these edits with quantitative measures of viral suppression and T-cell recovery in vivo. Results would have lasting effects in

HIV-1 therapy with CRISPR-Cas9 potentially replacing current long-term treatment methods like HAART to improve the quality of life, increase patient life span, and reduce adverse effects.

Methodology:

Human HSCs that already have the CCR5 gene will be retrieved from peripheral blood donors for the creation of humanized murine models. Fourteen neonatal NOD, B6.SCID Il2ry^{-/-} Kit^{W41/W41} mice aged 1 to 3 days will be used as they show a robust engraftment of long-term human HSCs, are highly immunodeficient, and do not require pre-conditioning irradiation.^{8,9} The mice will then be divided into a control group (n=7) and experimental group (n=7), as demonstrated in Figure 1. In the control group, 1×10^6 donor HSCs will be injected intrahepatically into the mice to achieve successful humanization of the mice.^{10,11} For the experimental group, 1×10^6 donor HSCs will be modified using CRISPR-Cas9 to induce the homozygous CCR5 Δ 32 mutation before being injected intrahepatically into the mice through a sterile syringe. Both groups will then be monitored for >10 T-cells/microliter of blood, allowing for an accurate model of human infection, before injection with HIV-1.¹²

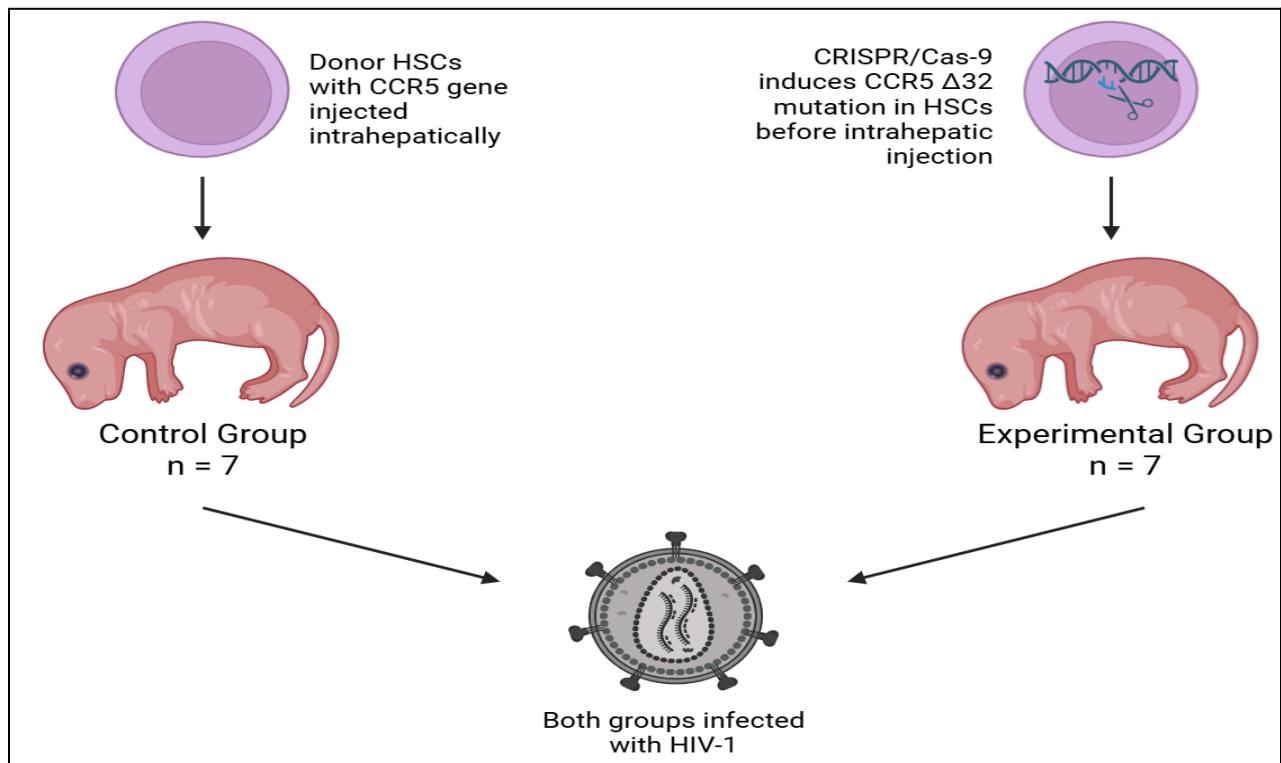


Figure 1: Procedure prior to HIV-1 infection. Donor HSCs will undergo modification using CRISPR-Cas9 in experimental groups, but remain unmodified in control groups, before being infected with HIV-1.

Post-infection procedures with HIV-1-positive mice will last 5 months to replicate chronic infection in humans and to collect enough data; previous studies with different humanized mice also sustained viral load for over 20 weeks.¹³ Mice will be bled every other week from the submandibular vein to draw blood to monitor the frequency of HIV-1-infected T-cells and viral RNA using flow cytometry and quantitative PCR, respectively, in both experimental groups.¹⁴ External variables including sex, diet, and environment must be uniform to prevent confounding results. Statistical procedure will include multiple Mann-Whitney U Tests, as sample sizes do not adhere to the Central Limit Theorem ($n < 30$) or normal distribution, and a Bonferroni correction, demonstrated in Figure 2.

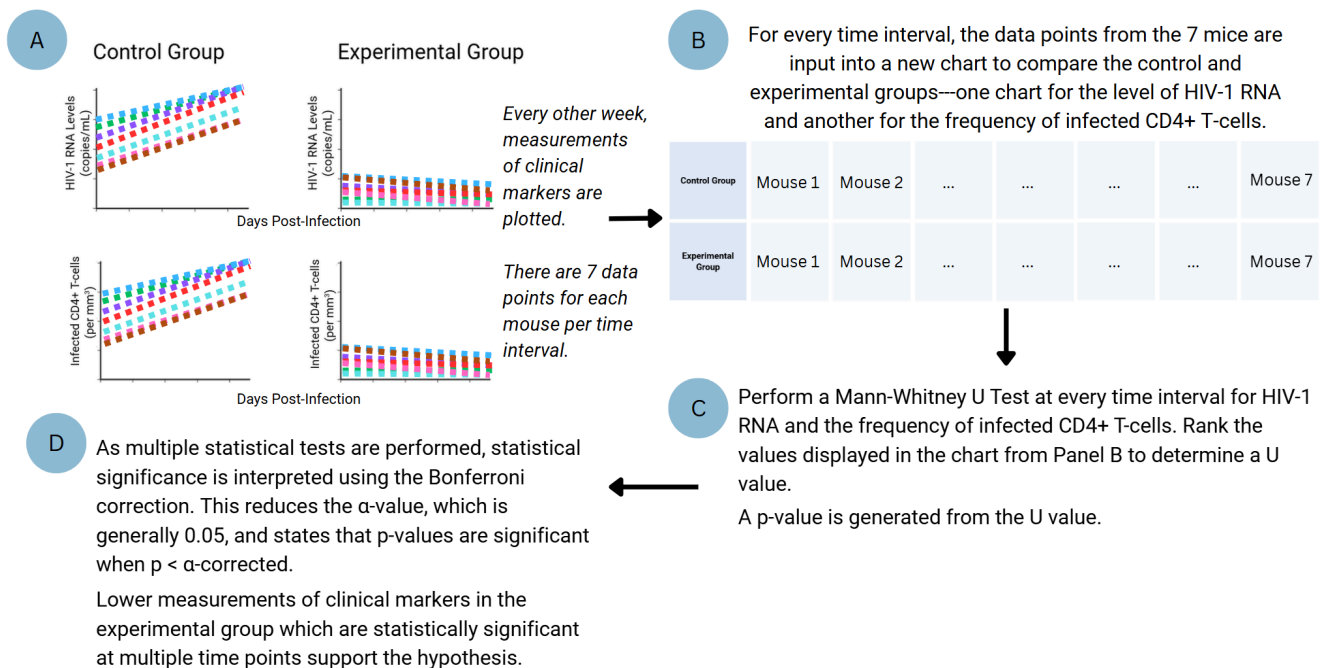


Figure 2: Statistical Procedures conducted using SPSS. Panel A displays sample data. Panel B organizes the measurements using a chart. Panel C and Panel D provide textual explanations.

Lower measurements of clinical markers in the experimental group, which are consistently significant, will support CRISPR-Cas9 efficiency.

Conclusion:

This study investigated the effectiveness of CRISPR-Cas9 in introducing the CCR5 Δ 32 mutation in HSCs to create humanized mice and determine HIV-1 resistance. Similar studies exist, but rely on random indel mutations: deletions/insertions of DNA bases introduced by cellular repair mechanisms following a CRISPR-Cas9 edit.^{3,15} Meanwhile, this study specifically targets the introduction of the CCR5 Δ 32 mutation, making results novel as it explores the application of a naturally occurring genetic mutation to defend against HIV-1.¹⁶ However, this study faces limitations. Incomplete editing efficiencies could lead to off-target effects, including the deletion of base pairs outside the CCR5 gene.¹⁷ Additionally, a small sample size may not deliver reliable results. Functions of the CCR5 co-receptor need to be investigated as multiple studies reported the CCR5 Δ 32 mutation increases risk for symptom severity in Influenza and the West Nile virus.¹⁸ Future studies should include larger sample sizes to test for a precise and safe delivery system, further validating the mutation's viability in non-human primates.

Definition Sheet:

- Human Immunodeficiency Virus Type 1 (HIV-1) = The most widespread type of HIV that primarily infects CD4+ T-cells. Over time, HIV-1 depletes these cells, weakening the immune response and making the body vulnerable to minor infections and certain cancers. If left untreated, HIV-1 leads to Acquired Immunodeficiency Syndrome (AIDS), characterized by severe immune suppression. Common symptoms include fever, swollen lymph glands, also called nodes, mainly on the neck, weight loss, and night sweats.^{19,20,21}
- Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)/Cas9 = Gene-editing technology derived from the adaptive immune system of prokaryotes. CRISPR sequences guide the Cas9 enzyme to a specific location in the genome to induce a double-stranded break. The process that follows allows insertion, deletion, or replacement of targeted genetic material.²²
- C-C Chemokine Receptor Type 5 (CCR5) = A protein expressed on the surface of white blood cells, including CD4+ T-cells. It is one of the main co-receptors that HIV-1 uses to enter and infect host cells.^{23,24}
- Hematopoietic Stem Cells (HSCs) = Stem cells that are found in the bone marrow and peripheral blood of humans. They can differentiate into all blood cell types, including immune cells such as T-cells, B-cells, and macrophages.²⁵
- Highly Active Antiretroviral Therapy (HAART) = A drug combination therapy that works by suppressing viral replication but not eliminating the virus. Lifelong adherence is required, and side effects, cost, and drug resistance are significant concerns.¹

- CD4⁺ T-Cells = A subset of lymphocytes that play a central role in coordinating immune responses. These cells are the primary targets of HIV-1. Loss of CD4⁺ cells due to viral infection leads to immune suppression.^{26,19}
- Berlin and London Patients = The first two known individuals functionally cured of HIV after receiving bone marrow transplants from donors with the homozygous CCR5 $\Delta 32$ mutation. These cases validated the CCR5-targeting strategy but also questioned the practicality and the ethicality of bone marrow transplants as a widespread solution.³
- NOD, B6.SCID Il2 $\gamma^{-/-}$ Kit^{W41/W41} Mice = A mouse strain ideal for engraftment of human hematopoietic cells.⁸

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